



Working with Time

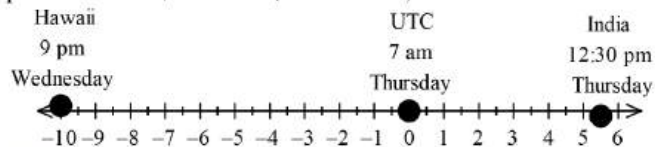
Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

- 1 The Coordinated Universal Time (UTC) of Hawaii = -10 hours. When it is 9 pm Wednesday in Hawaii, it is 12:30 pm Thursday in India. What is the UTC of India? 2

From 9 pm Wednesday to 12:30 pm Thursday is a difference of 15.5 hours.



Mapping the information around UTC 0 shows that India has a UTC of $+5.5$ hours

- 2 Melissa flies out of Brisbane at 11am on Sunday and arrives in Brussels 26 hours later. If Brussels is 9 hours behind Brisbane, find the time in Brussels when Melissa arrives? 4 am Monday 1
- 3 A plane takes 12 hours to fly from Adelaide (UTC $+9.5$) to Cape Town in South Africa (UTC $+2$). If Shaunna leaves Adelaide at 10:00 am, what is the local time when she arrives in Cape Town? 2

The time difference between the cities is $9.5 - 2 = 7.5$ hours.

Adelaide is ahead of Cape town.

Go back 7.5 hours from 10:00 am then add 12 hours flying time.

$10 \text{ am} - 7.5 \text{ hour} + 12 \text{ hours} = 2:30 \text{ pm}$.

It is 2:30 pm on the same day

- 4 Elizabeth lives in New York, USA (UTC -5) and Margaret lives in Sydney, NSW (UTC $+10$). Margaret makes a call to Elizabeth at 12:30 pm on Monday 24th February. February is a month when NSW has daylight saving time and the USA doesn't. What is the time in New York when Elizabeth receives the call? 1

*sydney is ahead
 $5 + 10 + 1$ (daylight savings)*

*$12 - 16 = -4$
which is 4 hours
before midnight
8:30 pm Sunday*

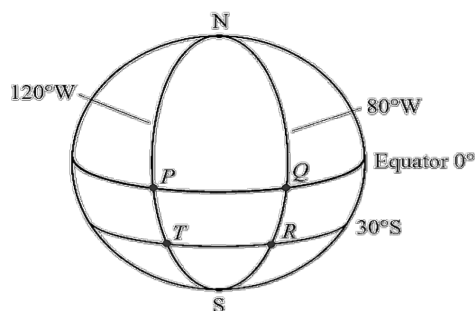
- 5 Cairo in Egypt is located at ($30^\circ N, 31^\circ E$). Sydney, in Australia, is located at ($33^\circ S, 151^\circ E$). Camilla lives in Sydney and needs to attend an online meeting with her business partners in Cairo starting at 11:00 am on Monday, local time in Cairo. What time and day, in Sydney, does Camilla need to log on to the meeting? Ignore daylight saving. 3

$$\text{difference in longitude} = 151^\circ - 31^\circ = 120^\circ$$

$$\text{time difference} = \frac{120^\circ}{15^\circ} = 8 \text{ hours} \quad (\text{since } 15^\circ = 1 \text{ hour})$$

$$\begin{aligned} \text{local time in Sydney} &= 11 \text{ am Monday} + 8 \text{ hours} \\ \text{when meeting starts} &= 7 \text{ pm Monday} \end{aligned}$$

- 6 Use the diagram below to name the points with these position coordinates:



- a. $30^{\circ}\text{S}, 120^{\circ}\text{W}$ T
 b. $0^{\circ}, 80^{\circ}\text{W}$ Q
 c. $30^{\circ}\text{S}, 80^{\circ}\text{W}$ R
 d. $0^{\circ}, 120^{\circ}\text{W}$ P

- 7 The coordinates of Tokyo are $(36^{\circ}\text{N}, 140^{\circ}\text{E})$ and the coordinates of Oslo are $(60^{\circ}\text{N}, 11^{\circ}\text{E})$. If the sun rises at 7 am in Oslo, what time will it be in Tokyo when the sun rises in Oslo? You may assume $15^{\circ} = 1 \text{ hour}$.

3:36 pm

- 8 Tokyo is one hour behind Sydney time and six hours ahead of Athens time. If it is 11 am in Sydney, what is the time in Athens?

A T S
4am 10am 11am
It is 4am in Athens

- 9 San Francisco is located at $(38^{\circ}\text{N}, 122^{\circ}\text{W})$. City X is located 40° to the South and 50° to the East of San Francisco. What are the coordinates of City X?

City X is located 40° to the South of 38°N .

latitude = $38^{\circ}\text{N} - 40^{\circ} = 2^{\circ}\text{S}$

City X is located 50° to the East of 122°W .

longitude = $122^{\circ}\text{W} - 50^{\circ} = 72^{\circ}\text{W}$

Therefore, the coordinates of City X are $(2^{\circ}\text{S}, 72^{\circ}\text{W})$.

- 10 Elizabeth lives in Sydney, NSW (UTC +10) and Margaret lives in Los Angeles, USA (UTC -8).

Margaret makes a call to Elizabeth at 12:30 pm on Monday 24th June. What is the date and time in Sydney when Elizabeth receives the call?

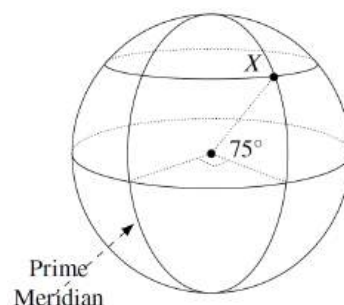
12:30
+ 18:00
30:30
- 24:00

6:30 am Tuesday 25th

- 11 The Coordinated Universal Time (UTC) of Sydney, Australia is +10 hours and the UTC of Mexico City, Mexico is -5 hours. What is the time in Mexico City if it is 8 am on Monday in Sydney?

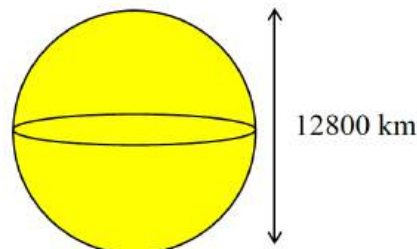
5 pm Sunday

- 12 What are the position coordinates of X in the diagram below?



- A. $75^{\circ}\text{S}, 90^{\circ}\text{E}$ B. $75^{\circ}\text{S}, 90^{\circ}\text{W}$
C. $75^{\circ}\text{N}, 90^{\circ}\text{E}$ D. $75^{\circ}\text{N}, 90^{\circ}\text{W}$

- 13 The earth is in the shape of a sphere with the diameter of 12 800 km.



- a. Show that the circumference of the earth along equator is approximately 40000 km?

$$C = \pi D = \pi \times 12800 \text{ km} \\ = 40221.239... \div 40000 \text{ km}$$

- b. A satellite is located 1000 km above sea level in space. How many km, to the nearest 100 km, does the satellite travel in each rotation?

$$C = \pi (12800 + 1000 \times 2) \\ = \pi \times 14800 = 46495.57.. \text{ km} \\ \div 46500 \text{ km}$$

- c. City A is located at $(35^\circ N, 125^\circ E)$. City B is located 60° to the east and 40° to the south of city A. What is the latitude and longitude of city B?

$$\begin{array}{c} 35^\circ N \\ -40 \\ \hline -5^\circ N \\ 125^\circ E \\ +60 \\ \hline 185^\circ E \end{array}$$

city B is at $(-5^\circ N, 185^\circ E)$

$$\therefore \text{city B is at } (5^\circ S, 175^\circ W)$$

- 14 Phoebe wants to catch a train from Parramatta to Blacktown. This is the timetable she would use. What is the latest train Phoebe can catch from Parramatta to arrive at Blacktown station before 7:15 am?

6:52 am

Central	06:10	06:23	06:23	06:25	06:34	-	06:38	06:40	06:53
Redfern	06:12	-	06:25	06:27	06:36	-	06:40	06:42	-
Burwood	-	-	06:45	-	-	-	07:00	-	-
Strathfield	06:24	06:36	06:48	06:39	06:48	-	07:03	06:54	07:06
Lidcombe	06:30	-	06:56	06:45	-	-	07:11	07:00	-
Auburn	-	-	06:59	-	-	-	07:14	-	-
Clyde	-	-	07:02	-	-	-	07:17	-	-
Granville	-	-	07:03	-	-	-	07:18	-	-
Harris Park	-	-	07:06	-	-	07:01	07:21	-	-
Parramatta	06:37	06:48	07:08	06:52	07:00	07:06	07:23	07:07	07:18
Westmead	06:40	-	-	06:55	07:03	07:09	-	07:10	-
Wentworthville	-	-	-	-	07:05	07:11	-	-	-
Pendle Hill	-	-	-	-	07:08	07:14	-	-	-
Toongabbie	-	-	-	-	07:10	07:16	-	-	-
Seven Hills	06:47	-	-	07:02	07:13	07:19	-	07:16	-
Blacktown	06:51	06:57	-	07:06	07:17	07:23	-	07:20	07:27

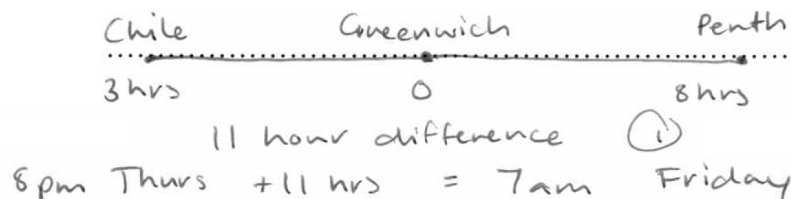
- 15 Pat lives in Sydney (UTC +10) and Freya lives in Seattle (UTC -8). Determine the time and day in Sydney if it is 10:00 pm on Thursday at Seattle.

4:00 pm Friday

- 16 A football match begins at 8:00pm Friday local time in New York, which has a Coordinated Universal Time (UTC) of -5. Rowan lives in Perth (UTC +8), if he wants to watch the match, what time will he need to turn on his television?

9:00am Saturday

- 17 Perth in Western Australia is 8 hours ahead of Greenwich in England. Santiago in Chile is 3 hours behind Greenwich. What is the day and time in Perth when it is 8pm on Thursday in Santiago?



- 18 Emma is in Sydney ($34^{\circ}S, 150^{\circ}E$) and is flying to Singapore ($1^{\circ}N, 105^{\circ}E$). Her plane is scheduled to leave Sydney at 9:45pm on Friday (Sydney time). The total flight time is 8 hours and 30 minutes. What day and time will it be in Singapore when Emma arrives?

Difference in longitude : $150 - 105 = 45^{\circ}$

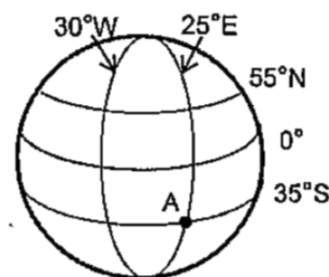
Time difference : $\frac{45}{3} = 3$ hours

The plane departs Sydney at 6:45 pm (Singapore time) and takes 8.5 hours

So the plane lands in Singapore at 3:15 am on Saturday (Singapore time)

- 19 What is the position of town A on the Earth's surface?

$(35^{\circ}S, 25^{\circ}E)$





Working with Time

Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 Elizabeth lives in New York, USA (UTC -5) and Margaret lives in Sydney, NSW (UTC +10). 2

Margaret makes a call to Elizabeth at 12:30 pm on Monday 24th February. February is a month when NSW has daylight saving time and the USA doesn't. What is the time in New York when Elizabeth receives the call?

The time difference between the cities is $10 + 5 = 15$ hours.
Sydney is ahead of New York, being plus (east) of NY.
When it is 12:30 daylight saving time, it is 11:30 Eastern Standard Time
So, go back 15 hours from 11:30 am Monday.
Back 12 hours is 11:30 pm Sunday.
Back a further 3 hours is 8:30 pm Sunday.

- 2 Gaby boards a plane in New York at 6 am on Wednesday 7th August, flying to Sydney. Sydney's time zone is UTC +10 and New York's is UTC-5.

- a. What is the time difference between the two cities? 1

Time difference = $10 + 5 = 15$ hours
So Sydney is 15 hours ahead.

- b. As she boards the plane, she phones her dad in Sydney to let him know she is leaving. 1

What is the time in Sydney when he gets the call?

Sydney is 15 hours ahead of 6 am Wednesday 7th.
6 hours ahead of 6 am is 12 pm Wed.
9 hours more ahead is **9 pm on Wednesday 7th.**

- c. If the flight takes 20 hours from time of boarding, what is the local time when she disembarks in Sydney? 1

The time she leaves New York (Sydney time) is 9 pm Wednesday.
Time for flight is 20 hours, so 3 hours to midnight and 17 hours after midnight so arrives at 5 pm Thursday 8th.

- 3 Cassie flew from London to Manila. Manila is 8 hours ahead of London. Her plane left London at 9.30 am Monday (London time), stopped for 5 hours in Singapore and arrived in Manila at 4.00 pm Tuesday (Manila time). What was the total flying time? (Ignore time zones.) 2

$$\begin{aligned}\therefore \text{Flight time} &= 9:30 \text{ (Mon) to } 4:00 \text{ pm (Tues)} \\ &\quad - (\text{stopover} + \text{time difference}) \\ &= 14.5 + 16 - (5 + 8) = 17.5 \text{ hours}\end{aligned}$$

- 4 Belinda lives in Dubbo, NSW, which is in the Eastern Standard Time zone of UTC + 10. Felicity lives in Denver, Colorado which is in the time zone of UTC− 7.

a. What is the time difference between Dubbo and Denver? (Include which city is ahead, in your answer). 17 hours difference, Dubbo ahead.

b. If Belinda makes a call to Felicity at 7.30 am on Friday 5th July (Dubbo time), what will be the local date and time in Denver when Felicity receives the call?

Denver = 17 hours behind 7:30am on Fri 5th July
7:30am Friday
✓ -12 → 7:30pm Thursday
-5 → 2:30pm Thursday
∴ 2:30pm Thursday 4th July in Denver

5 Shanghai in China has a UTC of +8 and Caracas in Venezuela has a UTC of −4. What is the day and time is it in Caracas when it is 10 am, Monday in Shanghai? 10 pm, Sunday

6 James lives in Marrickville and wants to take his children to Taronga Zoo. James needs to catch a train to Circular Quay and from there, catch the ferry to Taronga Zoo.

This is a part of the train timetable he would use.

Campsie	10:04	10:09	10:19	10:34	10:39	10:49	11:04
Canterbury	10:06	—	10:21	10:36	—	10:51	11:06
Hurlstone Park	10:09	—	10:24	10:39	—	10:54	11:09
Dulwich Hill	10:11	—	10:26	10:41	—	10:56	11:11
Marrickville	10:13	10:16	10:28	10:43	10:46	10:58	11:13
Sydenham	10:16	10:19	10:31	10:46	10:49	11:01	11:16
St Peters	10:19	—	10:34	10:49	—	11:04	11:19
Erskineville	10:21	—	10:36	10:51	—	11:06	11:21
Redfern	10:23	10:26	10:38	10:53	10:56	11:08	11:23
Central	10:27	10:30	10:42	10:57	11:00	11:12	11:27
Town Hall	10:30	10:33	10:45	11:00	11:03	11:15	11:30
Wynyard	10:33	10:36	10:48	11:03	11:06	11:18	11:33
Circular Quay	10:36	10:39	10:51	11:06	11:09	11:21	11:36
St James	10:39	10:42	10:54	11:09	11:12	11:24	11:39
Museum	10:41	10:44	10:56	11:11	11:14	11:26	11:41

This is a part of the ferry timetable he would use.

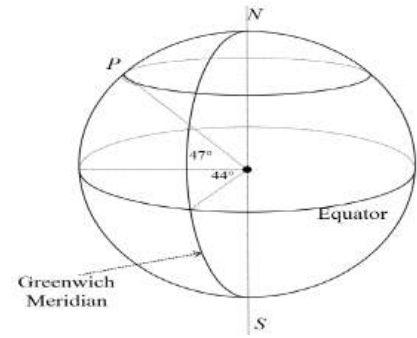
Monday to Friday	8	8	8	8	8	8	8
Circular Quay	08:50	09:20	09:50	10:20	10:50	11:20	11:50
Taronga Zoo	09:02	09:32	10:02	10:32	11:02	11:32	12:02
Mosman Bay	—	—	—	—	—	—	—

If James wants to be at the zoo by 11:30 am, what is the time of the latest train he will need to catch? 10:16 am

7 Mel leaves Darwin (UTC +9.5) at 1.40 pm on Monday and arrives in Singapore (UTC +8) at 4.50 pm on Monday. How long is Mel's direct flight from Darwin to Singapore?

Mel left Darwin at 12.10 Singapore time
From 12.10 PM to 4.50 PM : 4.40min

- 8 Point Q has the same longitude as P but is 60° further south.
 Point R has the same latitude as P and has a longitude of $61^\circ E$.
 Point P is in the time zone UTC+3.



- a. Write down the coordinates of P and Q .

$$P(47^\circ N, 44^\circ W), Q(13^\circ S, 44^\circ W)$$

- b. If point R is in the time zone UTC-4 what is the time

difference between P and R ?

7 hours

using diagram - R is 7 hours ahead of P

using text in question R is 7 hours behind P .

- 9 Yerevan in Armenia is 3 hours ahead of Lagos in Nigeria. Reykjavik in Iceland is 4 hours behind Yerevan. What is the time in Reykjavik when it is 3 pm in Lagos?

2 pm

- 10 Time zone and location details for two cities are given in the table below.

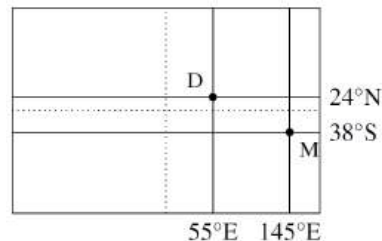
	Melbourne	Dubai
Latitude	$38^\circ S$	$24^\circ N$
Longitude	$145^\circ E$	$55^\circ E$
Time Zone	UTC+10	UTC+4

- a. Calculate the difference in latitude between Melbourne and Dubai.

$$\text{Difference in latitude} = 38 + 24 = 62^\circ$$

- b. A plane leaves Melbourne on Friday at 11:30 pm. The flight time to Dubai is 15 hours.

What will be the time and the day in Dubai when the plane is due to land?



Time difference = $10 - 4 = 6$ hours

Time of flight = 15 hours

	Melbourne	Dubai
Time of departure	11:30 pm Fri	5:30 pm Fri
Time of arrival		8:30 am Sat

$\therefore$ The flight arrives in Dubai at 8:30 am on Saturday morning.

- 11 If it is 10:35 am in Vancouver (UTC -8) what is the local time in Paris (UTC +1)?

A. 1:35 am

B. 5:35 pm

C. 3:35 am

D. 7:35 pm

- 12 Melbourne time is 6 hours ahead of Dubai time. A plane leaves Melbourne on Friday at 11.30 pm. The flight time to Dubai is 15 hours. What will be the time and the day in Dubai when the plane is due to land?

Arrival time = $11:30 + 15$ hours = 14:30 (Melbourne time)

$\therefore$ Arrival time in Dubai time = $14:30 - 6$ hours = 8:30 am on Saturday

13 Bert is in Moscow, which is three hours behind of Coordinated Universal Time (UTC). Karen is in Sydney, which is eleven hours ahead of UTC.

- a. Bert is going to ring Karen at 9 pm on Tuesday, Moscow time. What day and time will it be in Sydney when he rings? 1

Bert is 14 hours behind Karen. At 9 pm in Moscow:

Time in Sydney = 9 pm + 14 hours = 11 am Wednesday

- b. Karen is going to fly from Sydney to Moscow. Her flight will leave on Wednesday at 8 am, Sydney time, and will take 15 hours. What day and time will it be in Moscow when she arrives? 2

Arrival time = 8 am + 15 hours = 11 pm Wednesday (Sydney time)

$\therefore$ Time in Moscow = 11 pm less 14 hours = 9 am Wednesday

14 Karin is in Athens, which is two hours ahead of Greenwich Mean Time. Marco is in New York, which is five hours behind Greenwich Mean Time.

- a. Karin is going to ring Marco at 10 pm on Tuesday, Athens time. What day and time will it be in New York when she rings? 1

Athens is Greenwich + 2 hours

New York is Greenwich - 5 hours

Athens is 7hrs ahead of New York

$\Rightarrow$ 10pm Tuesday in Athens = 3pm Tuesday in New York

$\therefore$ It will be 3pm Tuesday when Karin rings

- b. Marco is going to fly from New York to Athens. His flight will leave on Wednesday at 9 am, New York time, and will take 11 hours. What day and time will it be in Athens when he arrives? 2

Flight takes 11 hours

If Marco leaves on Wed at 9am

$\Rightarrow$ He arrives at 9am + 11 hrs + 7 hours

$\therefore$ Marco arrives on Thurs at 3am